Experiment No. 5

Stack operations

```
#include <stdio.h>

int MAX = 5;
int stack[5];
int top = -1;

void push()
{
   int item;

   if(top == MAX - 1)
   {
      printf("Stack Overflow\n");
   }
   else
   {
      printf("Enter element to push: ");
      scanf("%d", &item);

      top = top + 1;
      stack[top] = item;

      printf("%d inserted into stack\n", item);
   }
}

void pop()
{
   int item;

   if(top == -1)
   {
      printf("Stack Underflow\n");
   }
   else
   {
      item = stack[top];
      top = top - 1;

      printf("Deleted element: %d\n", item);
   }
}

void display()
{
   int i;
```

```
    if(top == -1)
    {
        printf("Stack is empty\n");
    }
    else
    {
        printf("Stack elements are:\n");
        for(i = top; i >= 0; i--)
        {
            printf("%d\n", stack[i]);
        }
    }
}

int main()
{
    int choice;

    while(1)
    {
        printf("\n1. PUSH\n2. POP\n3. DISPLAY\n4. EXIT\n");
        printf("Enter your choice: ");
        scanf("%d", &choice);

        switch(choice)
        {
            case 1: push(); break;
            case 2: pop(); break;
            case 3: display(); break;
            case 4: return 0;
            default: printf("Invalid choice\n");
        }
    }
}
```

```
1. PUSH
2. POP
3. DISPLAY
4. EXIT
Enter your choice: 66
Invalid choice

1. PUSH
2. POP
3. DISPLAY
4. EXIT
Enter your choice: 1
Stack Overflow

1. PUSH
2. POP
3. DISPLAY
4. EXIT
Enter your choice: 3
Stack elements are:
88
56
76
54
23

1. PUSH
2. POP
3. DISPLAY
4. EXIT
Enter your choice: 4

c:\Users\PRASANNA\Desktop\DS practicals\5
```

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C:\Users\PRASANNA\Desktop\DS practicals\5th
acticals\5th practical\"stack_queue

1. PUSH
2. POP
3. DISPLAY
4. EXIT
Enter your choice: 2
Stack Underflow

1. PUSH
2. POP
3. DISPLAY
4. EXIT
Enter your choice: 1
Enter element to push: 23
23 inserted into stack

1. PUSH
2. POP
3. DISPLAY
4. EXIT
Enter your choice: 1
Enter element to push: 54
54 inserted into stack

1. PUSH
2. POP
3. DISPLAY
4. EXIT
Enter your choice: 1
Enter element to push: 76
76 inserted into stack

1. PUSH
2. POP
3. DISPLAY
4. EXIT
Enter your choice: 1
Enter element to push: 56
56 inserted into stack
```